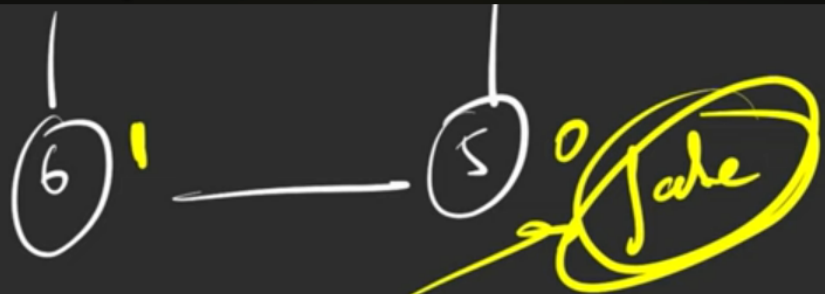


2.00



DFS(1, 0)
false
DFS(2, 1)
false
DFS(3, 0)
false
DFS(4, 1)
DFS(7, 0)
DFS(8, 1)
DFS(5, 0)
false
DFS(6, 1)

4 → {3, 7, 1}
5 → {4, 6}
6 → {2, 5}
7 → {4, 8}
8 → {7}

Start = 1

checking for 2



G-18. Bipartite Graph | DFS | C++ | Java

2.00

Bipartite Graph

Medium Accuracy: 40.1% Submissions: 79088 Points: 4

Given an adjacency list of a graph **adj** of V no. of vertices having 0 based index. Check whether the graph is bipartite or not.

Example 1:

Input:

0

Output Window

Compilation Results Custom Input

Problem Solved Successfully

You get marks only for the first correct submission if you solve the problem without viewing the full solution.

Test Cases Passed:

825 / 825

Your Total Score:

359

Total Time Taken:

0.47

Correct Submission Count:

Practice

C++ (g++ 5.4)

```
1 // } Driver Code Ends
2
3 class Solution {
4 private:
5     bool dfs(int node, int col, int color[], vector<int> adj[]) {
6         color[node] = col;
7
8         for(auto it : adj[node]) {
9             if(color[it] == -1) {
10                 if(dfs(it, !col, color, adj) == false) return false;
11             }
12             else if(color[it] == col) {
13                 return false;
14             }
15         }
16         return true;
17     }
18
19 public:
20     bool isBipartite(int V, vector<int> adj[]) {
21         int color[V];
22         for(int i = 0; i < V; i++) color[i] = -1;
23
24         for(int i = 0; i < V; i++) {
25             if(color[i] == -1) {
26                 if(dfs(i, 0, color, adj) == false) return false;
27             }
28         }
29         return true;
30     }
31 };
32 // } Driver Code Ends
```

Compile & Run

14:01 / 14:53